Lecture 4

Random Variables

Please have a coin handy that you can toss during the lecture,
or use `sample(c("Heads","Tails"),1)` in R

Probability

In 2022, Oz Lotto changed to drawing 7 balls from 47.

What is the probability of winning the 1st division prize (all 7 numbers)?

Poll Question

Suppose we pick a student at random from the class.

What is the probability that they are 170 cm tall?

1. 0.000
2. 0.061
3. 0.162
4. 0.633

Key Concepts

- **population**: a complete set of individuals or objects we want information about
- **sample**: a subset of the population

- **parameter**: a numerical characteristic of a population
- **statistic**: a numerical characteristic of a sample

Key Concepts

A **random process** (or experiment) is a repeatable procedure with uncertain results.

- **outcome**: a result of a random process
- **sample space**: the set of all possible outcomes
- **event**: a subset of a sample space

An experiment involve tossing a coin three times. Let A be the event that tails occur more often than heads. What is the sample space? What is A?

Key Concepts

Probability is a numerical measure of the likelihood that a specific event will occur.

Probability is a real-valued function P that assigns a positive number to each event A in a sample space Ω, satisfying the followimg axioms:

(i) $\mathsf{P}(A) \geq 0$

(ii) $\mathsf{P}(\Omega) = 1$

(iii) $\mathsf{P}(A_1 \cup A_2 \cup \ldots) = \mathsf{P}(A_1) + \mathsf{P}(A_2) + \ldots$ for disjoint events $A_1, A_2, \ldots$

Conditional Probability

$\mathsf{P}(A|B)$ denotes the conditional probability of A occurring given B has occurred, defined as

$$\mathsf{P}(A|B) = \frac{\mathsf{P}(A \cap B)}{\mathsf{P}(B)}.$$

Independence

We say that two events, A and B, are *independent* if

$$\mathsf{P}(A \cap B) = \mathsf{P}(A)\,\mathsf{P}(B).$$

Tossing a Coin

Use a method to pick from two equally likely outcomes, ‘H’ or ‘T’:

- Toss a coin (Heads or Tails)
- Use `sample(c("Heads","Tails"),1)` in R
- Use the scientific calculator on your phone to pick a random number (≥ 0.5 for ‘H’ and < 0.5 for ‘T’)

Poll Question

Toss a coin twice and indicate your outcome.

1. HH
2. HT
3. TH
4. TT

Sensitive Questions

Randomness can provide perfect anonymity when answering sensitive questions.

The UQ Extend content describes one method for doing this.

Here we will use the original method proposed by Warner in 1965.

Warner's Method

- Toss a coin twice and note the outcomes without anyone else seeing
- If you had HH then answer the question *truthfully*
- If you had anything else then answer the *opposite of the truth*

Question from Bégin, G., Boivin, M. & Bellerose, J. (1979). Sensitive data collection through the random response technique: some improvements. *The Journal of Psychology*, **101**, 53–65.

Poll Question

Which question from the Bégin *et al.* paper you would like the class to answer using Warner's method?

APPENDIX: QUESTIONNAIRE

1. Are you a Catholic?
2. Do you approve of premarital sex for engaged couples?
3. Do you approve of kissing on the first date?
4. Have you ever masturbated?
5. Have you ever had an homosexual experience?
6. Have you ever had a complete sexual intercourse?
7. Have you ever touched a partner's genitals with your mouth or have you ever had a partner touch your genitals with his (her) mouth?
8. Do you think high school students should have a course in sex education?
9. If, today, you or your girl friend would accidentally get pregnant, would you seriously consider the possibility of an abortion?

Poll Question

Use Warner's Method to respond to the chosen question.

- Toss a coin twice and note the outcomes without anyone else seeing
- If you had HH then answer the question *truthfully*
- If you had anything else then answer the *opposite of the truth*

1. Yes
2. No

Sensitive Questions

How do we estimate the population proportion?

Expected Value

The expected value $\mathsf{E}(X)$ of a *discrete* random variable X is given by

$$\mathsf{E}(X) = \mu = \sum \text{value} \times \text{probability} = \sum_x x\, \mathsf{P}(X = x)$$

For a *continuous* random variable, the expected value is given by

$$\mathsf{E}(X) = \mu = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

Expected Value

Example In 3-number Keno, you choose 3 numbers from the 80, winning \$44 if all 3 come up in the 20 chosen balls, \$1 if 2 come up and nothing otherwise.

It can be shown that the probability distribution of winnings is

x	0	1	44
$\mathrm{P}(X = x)$	0.8473	0.1388	0.0139

What is the expected winnings from a game of 3-number Keno?

Law of Large Numbers

The *law of large numbers* states that:

As the number of trials increases, sample proportions get closer to true probabilities and sample means get closer to expected values.

Be careful of the "law of small numbers", also called the *gambler's fallacy*:

"I've lost so many times in a row now, I must be due for a win"

Variance

We can quantify variability of a random variable X using the *expected squared deviation* about the mean, called the **variance**:

$$\mathsf{var}(X) = \sigma^2 = \mathsf{E}\left[(X - \mathsf{E}(X))^2\right].$$

- For discrete X, $\mathsf{var}(X) = \sum_x (x - \mathsf{E}(X))^2 \, \mathsf{P}(X = x)$.
- For continuous X, $\mathsf{var}(X) = \int_{-\infty}^{\infty} (x - \mathsf{E}(X))^2 f(x)\, dx$.

What is the variance of the winnings from a game of 3-number Keno?

Standard deviation

It is easier to interpret variability if it has the same scale as the original values. We use the **standard deviation**, the square root of the variance:

$$\mathsf{sd}(X) = \sqrt{\mathsf{var}(X)} = \sigma$$

If $\mathsf{sd}(X) = 0$, what can we say about X?

Combining and Transforming Random Variables

We are frequently interested in combining and transforming random variables to make a new random variable.

Rather than calculate them from scratch, there are general rules we can use to get the expected value and standard deviation of the new variable.

Poll Question

Let X be the winnings from 3-number Keno and suppose one day the casino gives you a bonus \$1 each time you play, giving a new random variable

$$Y = X + 1.$$

What are the expected value and standard deviation of Y?

1. $\mathsf{E}(Y) = 1.75$ and $\mathsf{sd}(Y) = 5.14$
2. $\mathsf{E}(Y) = 0.75$ and $\mathsf{sd}(Y) = 6.14$
3. $\mathsf{E}(Y) = 1.75$ and $\mathsf{sd}(Y) = 6.14$

Poll Question

Let X be the winnings from 3-number Keno and suppose the next day the casino doubles whatever you win, giving a new random variable

$$Y = 2X.$$

What are the expected value and standard deviation of Y?

1. $\mathsf{E}(Y) = 1.50$ and $\mathsf{sd}(Y) = 5.14$
2. $\mathsf{E}(Y) = 0.75$ and $\mathsf{sd}(Y) = 10.28$
3. $\mathsf{E}(Y) = 1.50$ and $\mathsf{sd}(Y) = 10.28$

Poll Question

Let X be the winnings from 3-number Keno and suppose that on the last day when you 'win' you give the casino \$1 or \$44, giving a new random variable

$$Y = -X.$$

What are the expected value and standard deviation of Y?

1. $\mathsf{E}(Y) = 0.75$ and $\mathsf{sd}(Y) = -5.14$
2. $\mathsf{E}(Y) = -0.75$ and $\mathsf{sd}(Y) = 5.14$
3. $\mathsf{E}(Y) = -0.75$ and $\mathsf{sd}(Y) = -5.14$

Two Random Variables

Suppose we now play two games of Keno.

What are the expected value and standard deviation of our *total* winnings?

What are the expected value and standard deviation of our *average* winnings per game?